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Adsorption of Alkyltrimethylammonium Bromides at the Various Interfaces

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The adsorption of alkyltrimethylammonium bromide homologues has been studied at oil/water and air/water interfaces. Dodecyltrimethylammonium bromide, tetradecyltrimethylammonium bromide, and hexadecyltrimethylammomium bromide were used. As an oil phase the aliphatic hydrocarbons of the same chain length as that of surfactants molecules were used. The interfacial tension was measured by the drop volume method and the equilibrium values were received. The surface tension was measured by the drop weight method. The values of surface excess Γ were calculated from the Gibbs equation. The experimental results were tested by the Langmuir isotherm as well as by the isotherm developed by E. Blomgren and J. O'M. Bockris (*J. Phys. Chem.* 63, 1475 (1959)).

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Key Words: **adsorption; A/W and O/W interfaces; alkyltrimethylammonium bromides.**

INTRODUCTION

The adsorption of surface active substances at the air/water interface has been widely investigated for various surfactants. Much less investigations have been devoted to the oil/water interface and only in few papers have the adsorption properties of particular compounds at the air/water and oil/water interfaces been compared.

Properties of adsorbed films of surface active substances at the oil/water interfaces have been investigated by many scientists. In most cases, however, they focused on the relation with micelle formation rather than on the adsorption itself (1, 2). It is well recognized that the interfacial properties of surfactants at the air/water surface are quite different to their properties at the oil/water interface (3–5). One would expect greatly increased adsorption at the oil/water interface (because of the hydrocarbon–surfactant interaction) compared with the extent of adsorption at the air/water surface.

Hutchinson (6) has suggested that oil molecules are present at the interface along with the adsorbed surfactants molecules and that a competition exists at the interface between the nonpolar portion of the surfactant and oil.

Aveyard *et al.* (7) have found that for close-packed monolayers at alkane/water interfaces the area per surfactant molecule is significantly dependent on alkane chain length as a result of the stronger penetration of the shorter alkanes into the chain side of the surfactant monolayers. They have stated that the alkane penetration stabilizes microemulsion droplets in Winsor systems.

It is well recognized (8) that there exists cohesion between carbon chains of surfactant molecules at the air/water interface. However, Boucher *et al.* (9) implies that at the oil/water interface such cohesion is absent; thus at such interfaces one may expect an adsorption of surfactants smaller than that at the air/water surface.

Analyzing the results presented in the literature one can state that the extent of adsorption at the different interfaces relates to the bulk concentration of surfactants in water.

Gilap *et al.* [4] have found that the adsorption of sodium dodecyl sulfate (SDDS) and sodium decyl sulfate (SDS) at the oil/water interface is higher than at the air/water surface, and the ratio of $\Gamma_{o/w}$ to $\Gamma_{a/w}$ equals 13 for SDDS and 35 for SDS. They implied that in the investigated region of concentration below 2×10^{-4} cmc (the region of the ideal adsorption), the surfactant molecules extend into the oil phase and are aligned as a result of the interactions between the oil molecules and the alkyl chains of the surfactant.

Molecular interactions at the interface can be analyzed on the basis of the deviation from the ideal adsorption. There are numerous papers dealing with nonideality of adsorption of various surfactants. Mostly, the Frumkin isotherm is applied for this purpose (10–13). However, on the basis of the Blomgren and Bockris equation (14) the explanation of the origin of intermolecular interactions is possible (15, 16).

The aim of this work is to examine the extent of deviation from ideal adsorption of alkyltrimethylammonium bromides at the air/water and water/hydrocarbon interfaces.

THEORY

The Langmuir isotherm of adsorption describes an ideal adsorption process (in sense of zero interaction between the adsorbed species). This isotherm can be used to calculate standard free energy of adsorption, $-\Delta G^{o}$,

$$\frac{\Theta}{1-\Theta} = \frac{f_A c_A}{55.5} \exp\left(-\frac{\Delta G^{o}}{RT}\right), \qquad [1]$$

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where Θ is the interface coverage, C_A is a bulk concentration of surfactant A, f_A is an activity coefficient for A in bulk, and 55.5 is a molecular concentration of water.

In real systems deviations from the ideal adsorption occur. These deviations originate from the intermolecular interactions of the adsorbate molecules at the interface. The energy of these interactions in the adsorbed layer, U_{AA} may be expressed by

$$U_{AA} = RT \ln f_A^{\mathrm{a}}, \qquad [2]$$

where f_A^{a} is an activity coefficient for A in the adsorbed state.

Taking into account these interactions Blomgren and Bockris (8) obtained the isotherm equation

$$\frac{\Theta}{1-\Theta} = \frac{f_A c_A}{55.5} \exp\left[-\frac{1}{RT}\left(\Delta G_0^{\mathrm{o}} + U_{AA}\right)\right], \qquad [3]$$

where $-\Delta G_0^{\mathrm{o}}$ is the standard free energy of adsorption corresponding to $\Theta = 0$.

The energy of the surface-active ion interactions depends mainly on the coulombic repulsion and on the Van der Waals attractive forces, and may be calculated from

$$U_{AA} = N\left(\frac{e_0^2}{\varepsilon_{\mathrm{s}} r} - \frac{3h\nu\alpha^2}{4\varepsilon_0 r^6}\right), \qquad [4]$$

where N is Avogadro's number, e_0 is elementary charge, ε_{s} is the static dielectric constant of solvent in the adsorbed layer, ε_0 is the electronic dielectric constant of the solvent in the adsorbed layer, ν is a characteristic frequency of the electronic oscillators in the adsorbate molecules, α is the electronic polarizability of the adsorbate, and r is the average distance between the adsorbed centers.

Introducing

$$\alpha = R^2 \qquad [5]$$

$$r = \sqrt{\frac{S}{\Theta}}, \qquad [6]$$

where S is the cross-sectional molecular area and equals ΠR^2, where R is the "radius" of the ionic group, then Eq. [4] may be expressed as

$$U_{AA} = N\left(\frac{e_0^2\sqrt{\Theta}}{\varepsilon_{\mathrm{s}}\sqrt{S}} - \frac{3h\nu\alpha^2}{4\varepsilon_0\pi^3}\Theta^3\right). \qquad [7]$$

The first term in this equation corresponds to the electrostatic repulsion; the second term relates to the dispersive forces. At low surface coverages Θ, it means at great distances between adsorbing molecules, the coulombic forces (propotional to the value $1/r$) overcome the dispersive attraction (proportional to the value $1/r^6$); thus the standard free energy of adsorption decreases with the increase of Θ. At great coverage Θ, the dispersive attraction is greater than coulombic repulsion, which causes the increase of the standard free energy of adsorption with the increase of Θ.

Thus, the $-\Delta G^{\mathrm{o}} = f(\Theta)$ plot should pass through a minimum, and at very low Θ, where coulombic forces predominate, the relation $-\Delta G^{\mathrm{o}} = f(\Theta^{1/2})$ should be linear. Such relations were observed for aromatic amines at the water/mercury interface (14), and for alkyl sulfates at the free surface of their aqueous solutions (15).

EXPERIMENTAL

The adsorption of alkyltrimethylammonium bromide homologues has been studied at oil/water and air/water interfaces. Dodecyltrimethylammonium bromide (DTAB), tetradecyltrimethylammonium bromide (TTAB), and hexadecyltrimethylammomium bromide (CTAB) were used. Surfactants were four times crystalized from acetone. There was no minimum on the surface tension isotherms near the cmc.

As an oil phase the aliphatic hydrocarbons of the same chain length as that in the surfactant molecules were used (dodecane, tetradecane, hexadecane). Hydrocarbons were purified by the following method: shaking with sulfuric acid, washing, distilling, and finally passing through an alumina column. Their purity was cheked on the basis of water/hydrocarbon interfacial tension values (51.90 mN/m for dodecane, 52.72 mN/m for tetradecane, and 52.78 mN/m for hexadecane).

Surface tension and interfacial tension were measured at 298 K using the drop-weight or drop-volume method, respectively. The accuracy of the methods was ± 0.1 and ± 0.25 mN/m, respectively. The equilibrium adsorption was achieved during the measuring time. The drop formation time ranged from 15 to 45 min, depending on surfactant concentration.

The results of the surface tension measurements are shown in Fig. 1 and the results of interfacial tension are shown in Figs. 2–4.

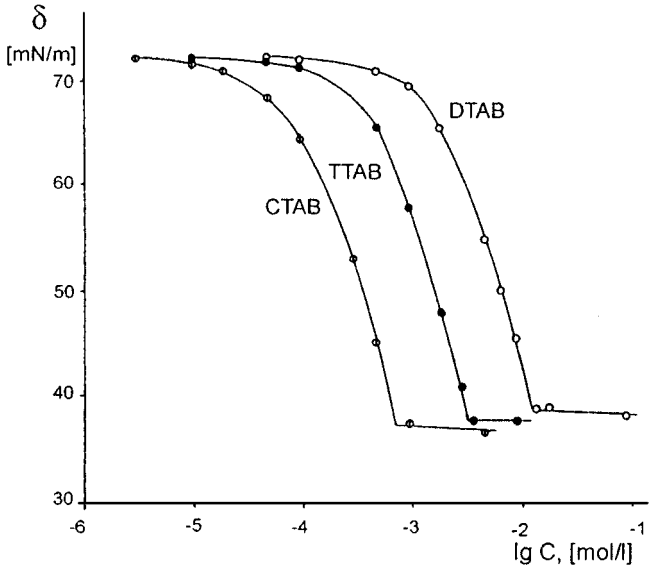


FIG. 1. Surface tension of aqueous solutions of alkyltrimethylammonium bromides versus surfactant concentration in water.

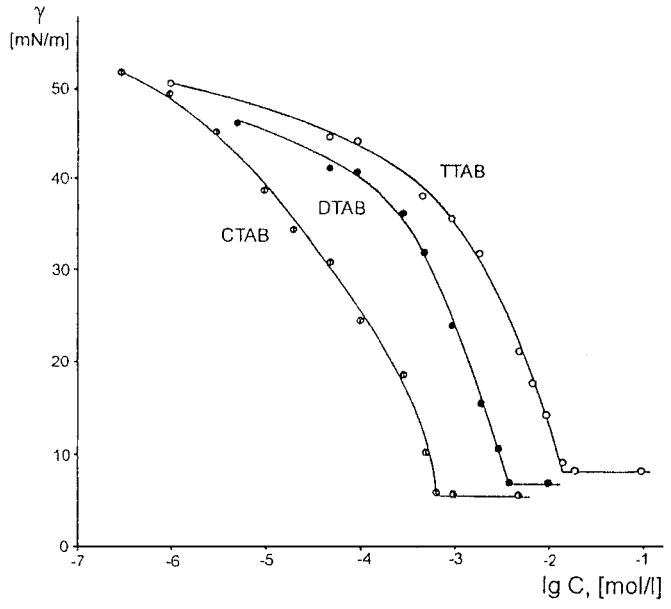


FIG. 2. Interfacial tension of alkyltrimethylammonium bromides at the dodecane/water interface.

The polynomials of the third degree were fitted to the experimental data over a range of surfactant concentration below the cmc. The surface excess Γ was calculated from the Gibbs equation by applying the fitted polynomials. The isotherms of DTAB, TTAB, and CTAB adsorption are presented in Figs. 5–7.

As can be seen from Figs. 5–7 the surface excess is lower at the A/W interface than at the O/W interface, which was expected and was also stated by Gillap *et al.* for sodium alkyl sulfates (4). This may be explained as a result of additional interaction

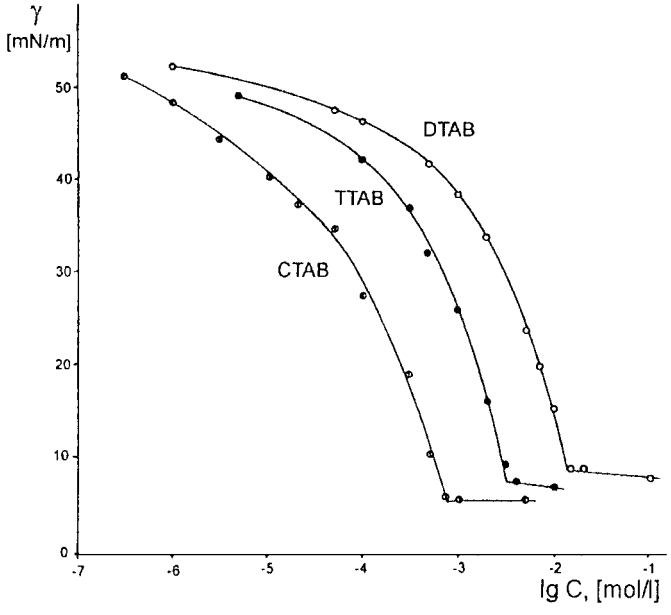


FIG. 3. Interfacial tension of alkyltrimethylammonium bromides at the tetradecane/water interface.

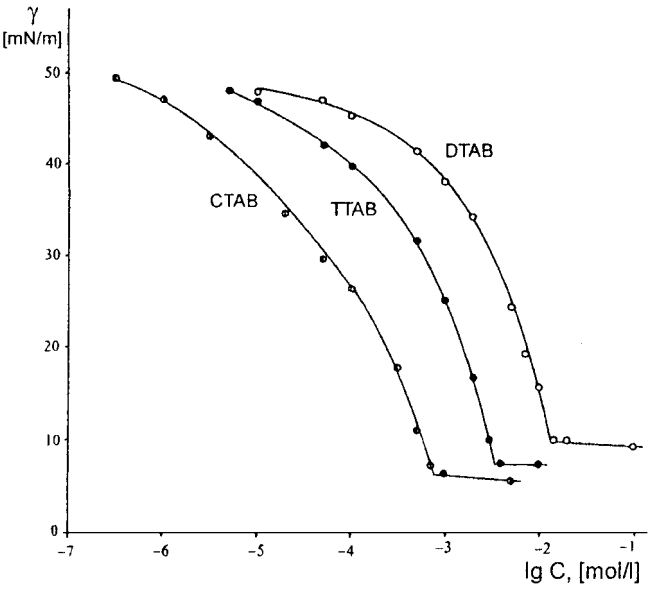


FIG. 4. Interfacial tension of alkyltrimethylammonium bromides at the hexadecane/water interface.

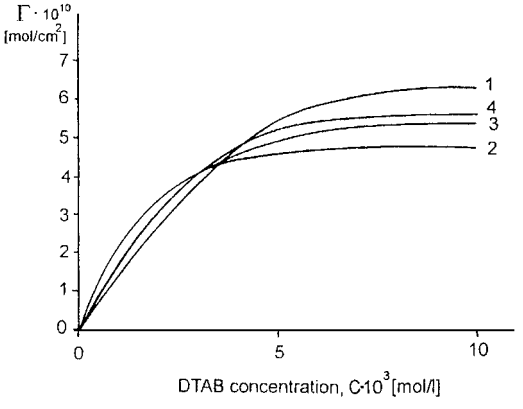


FIG. 5. Adsorption isotherm of dodecyltrimethylammonium bromide at various interfaces. Second phase (against water): 1, air; 2, dodecane; 3, tetradecane; 4, hexadecane.

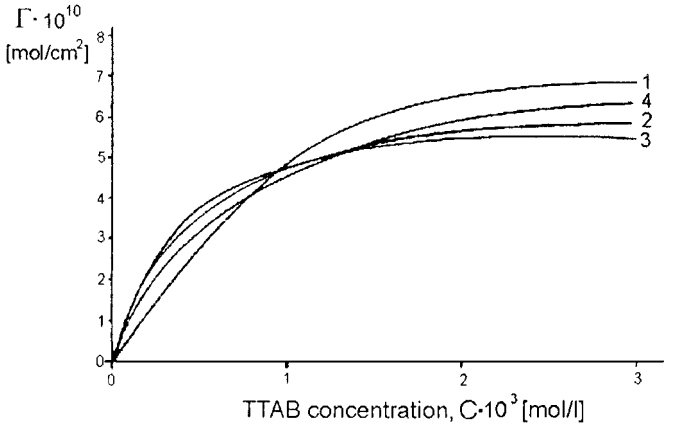


FIG. 6. Adsorption isotherm of tetradecyltrimethylammonium bromide at various interfaces. Remarks as in the legend to Fig. 5.

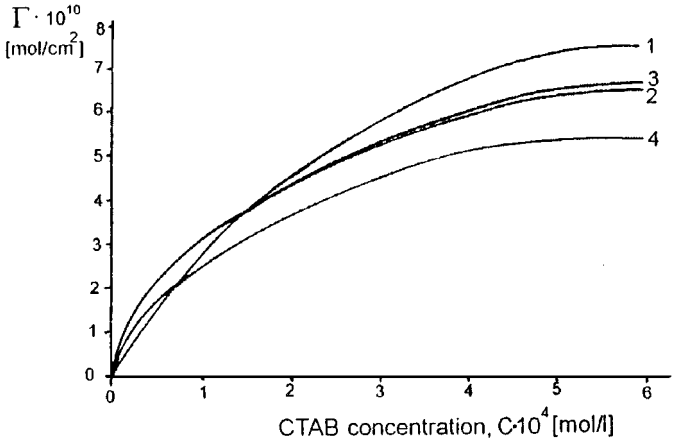


FIG. 7. Adsorption isotherm of hexadecyltrimethylammonium bromide at various interfaces. Remarks as in the legend to Fig. 5.

between oil molecules and alkyl chains of surfactants in the interfacial region, which does not take place at the A/W interface. However, this is true only at low Θ. At higher Θ, the surface excess is higher at the A/W interface than at the O/W interface, because in the last case both types of molecules contribute to the interfacial film and as a result, the number of adsorbed surfactant molecules is limited. This is not the case at the A/W interface, where only surfactant molecules assemble.

Analyzing the results presented in Figs. 5–7 one can state that the greatest interaction is observed in the case where the alkyl chains match, which results the lowest Γ_{max} for DTAB at the dodecane/water interface (Fig. 5, curve 2), for TTAB at the tetradecane/water interface (Fig. 6, curve 3), and for CTAB at the hexadecane/water interface (Fig. 7, curve 4).

DISCUSSION

Further interpretation of the experimental results for alkyltrimethylammonium bromides was based on the analysis of the $-\Delta G^o = f(\Theta)$ plots. Standard free energy of adsorption ΔG^o was calculated from Eq. [1] and the remarkable deviations of ΔG^o from constancy are seen to occur (Figs. 8–10). Thus it may be stated that the adsorption of RTMABr in the investigated region of concentration was far from ideal.

The shape of the curves was analyzed on the basis of interaction forces taken into consideration in the Blomgren and Bockris isotherm (14). This isotherm was used earlier for the study of interaction of fatty acid molecules at dodecane/water interface and their sodium salts at the air/water surface (16). It was also useful for estimation of dielectric constants of adsorbed monolayers of lauric and myristic sodium salts (17).

It was found that the standard free energy of adsorption of RTMABr is greater at the O/W interface than at the A/W interface within the whole range of Θ values; however, the observed differences are greater at low surface coverages Θ. Simulataneously, the greatest $-\Delta G^o$ values are observed for the interfaces of water against the hydrocarbon of which the chain length equals the alkyl chain length of RTMABr. For example, the highest $-\Delta G^o$ values for DTMABr was observed at the dodecane/water interface (Fig. 8), while for CTMABr at the hexadecane/water interface (Fig. 10).

It may be observed that the plots of $-\Delta G^o = f(\Theta)$ for the free surfaces have no minimum values, which was expected on the basis of the results for other surfactants. These curves (lines 1, Figs. 8–10) indicate that at low Θ values, the dispersive attraction is much greater or that coulombic repulsion is much lower than those predicted on the basis of Eq. [5]. Also the relations $-\Delta G^o = f(\Theta)$ are not linear at low Θ, which means that the

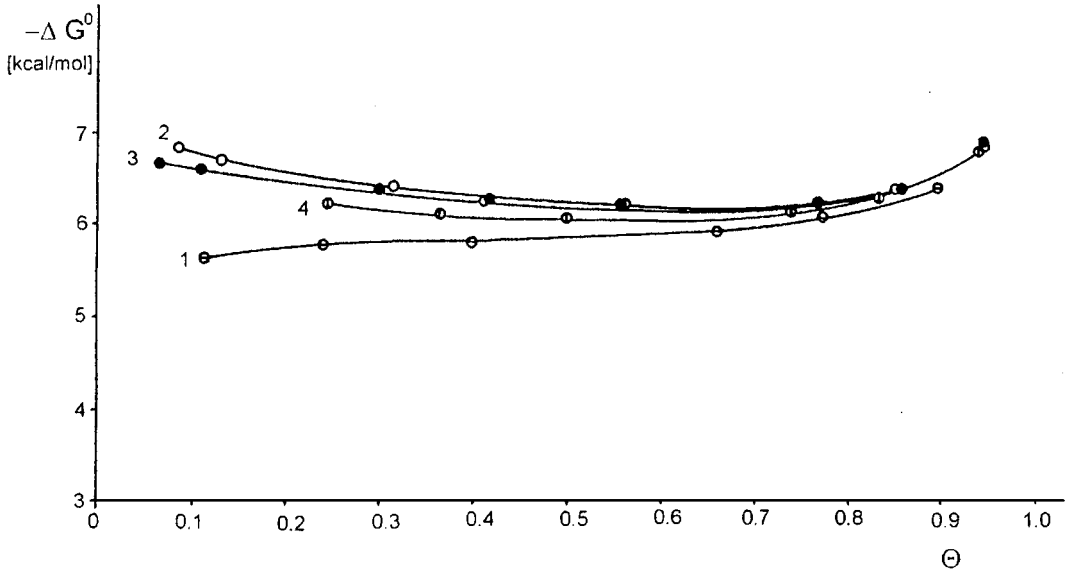


FIG. 8. Standard free energy of adsorption of dodecyltrimethylammonium bromide at various interfaces. Remarks as in the legend to Fig. 5.

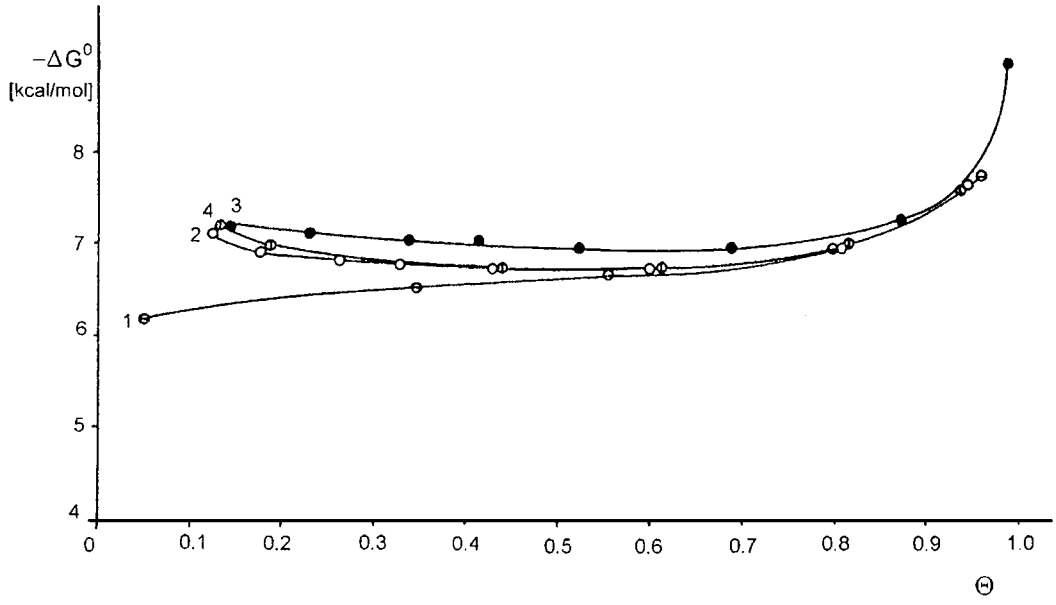


FIG. 9. Standard free energy of adsorption of tetradecyltrimethylammonium bromide at various interfaces. Remarks as in the legend to Fig. 5.

coulombic interactions are disturbed by the dispersive attraction, which was stronger than those expected from the results obtained for aromatic amines (14) and alkyl sulfates (15). The reason for such relation is probably the presence of three methyl groups surrounding the polar head in the RTMABr molecules. This diminishes electrostatic interactions between the polar groups. Simultaneously, the additional dispersive interaction between methyl groups of neighboring molecules increases the attraction of these molecules at the A/W interface.

In view of the above considerations the presence of minima on the curves $-\Delta G^{o} = f(\Theta)$ for hydrocarbon/water interfaces seems to be unexpected. The presence of these minima means that at low Θ values the dispersive forces at the hydrocarbon/water interfaces are lower than those at the air/water interface. This observation may be explained by the appearance of the additional dispersive forces between hydrocarbon molecules of the oil phase and the alkyl chains of the surfactants. These forces are strong and as a result the hydrocarbon molecules surround

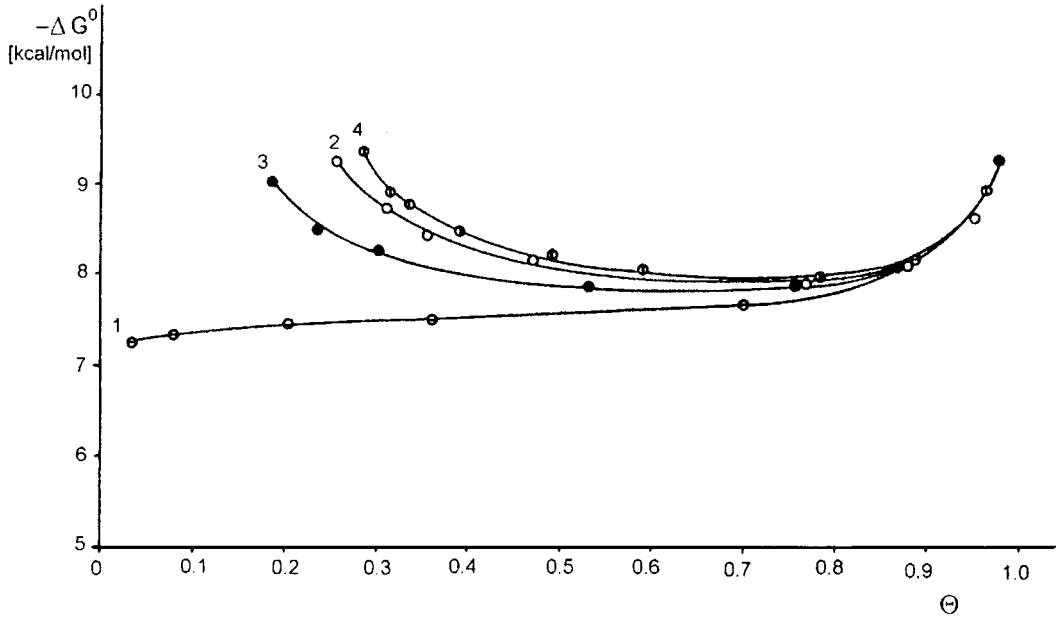


FIG. 10. Standard free energy of adsorption of hexadecyltrimethylammonium bromide at various interfaces. Remarks as in the legend to Fig. 5.

each molecule of surfactant adsorbed at the interface. Thus, the distances between adsorbed surfactant molecules are greater and the interactions of alkyl groups of surfactants are weaker than those at the free surface. So, at the low coverage Θ the coulombic repulsion overcomes dispersive forces, and a minimum on the curves $-\Delta G^{o} = f(\Theta)$ is observed.

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